

TECHNICAL GUIDE

Pilot & Scale-Up Solutions

Material-led testing, pilot validation and process data for a clearer path to production.

Laboratory screening | Pilot trials | Scale-up planning

<http://oltdryer/>



A practical bridge from material question to production decision

Spray drying scale-up starts with the material and the powder target. Each stage should answer a specific decision before the project moves forward.

01

Material review

Feed properties, available sample and powder target

02

Laboratory screening

Feasibility and an initial operating window

03

Pilot validation

Longer runs, recovery and process stability

04

Production direction

Capacity, configuration, utilities and controls

A successful laboratory test establishes direction. Pilot validation adds representative run time and a stronger basis for production planning.

Pilot testing focuses on the questions that short trials cannot answer



Representative operating behavior

- Process stability during longer runs
- Powder collection and wall deposition behavior
- Repeatability across operating conditions
- Practical cleaning and changeover access
- Data for equipment and auxiliary selection

Pilot work is especially useful when the feed is sticky, heat sensitive, viscous, solvent based or difficult to collect.

Six steps from sample submission to a documented next step

01

Submit sample

Material details and available quantity

02

Define scope

Goals, constraints and success criteria

03

Lab screening

Initial feasibility and data review

04

Pilot testing

Representative runs and observations

05

Review data

Compare conditions and powder results

06

Plan next step

Report, equipment direction or more testing

Test scope agreed before execution

Useful input for a productive first technical review

Share what you know

- Feed form and composition
- Solids content and viscosity when known
- Water, organic solvent or mixed solvent basis
- Thermal sensitivity and stability concerns
- Target moisture, particle size and powder behavior
- Available sample and expected production capacity

Unknown values can be clarified during engineering review.



Laboratory, pilot and industrial systems answer different questions



LABORATORY

1.5-3.0 L/h

Feasibility and formulation

Use limited material to identify a workable drying direction and observe initial powder behavior.

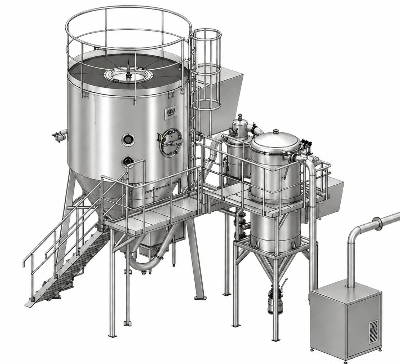


PILOT

5-30 L

Validation and optimization

Run longer trials to review stability, recovery and scale-up assumptions.



INDUSTRIAL

50-100+ L

Production integration

Configure the dryer, collection, controls and utilities around commercial duty.

A pilot configuration connects feed handling, drying and powder collection



1

Feed system

Stable delivery and suitable agitation

2

Atomization

Nozzle or centrifugal direction based on feed and particle target

3

Drying chamber

Airflow and thermal conditions for the required duty

4

Collection

Cyclone, filter or project-specific recovery direction

5

Controls

Monitored parameters and repeatable operating records

Pilot data records the relationships that matter for scale-up

FEED	Solids, viscosity, temperature and stability
ATOMIZATION	Method, pressure or speed and droplet behavior
THERMAL	Inlet and outlet conditions with product response
AIRFLOW	Drying balance, residence behavior and exhaust condition
POWDER	Moisture, particle properties, yield and collection

Final production design also depends on equipment geometry, utilities, controls and operating discipline.



A clear record for the next engineering decision

**Test report**

Process conditions, results and observations

Powder review

Available moisture, particle, morphology or flow data

Material balance

Yield and collection observations within the test scope

Operating direction

A practical process window for further validation

Scale-up plan

Suggested equipment scale and next technical step

Pilot evidence informs full-system engineering



Feed duty

Composition, solids and evaporation load

Process window

Validated temperature and atomization direction

Recovery

Cyclone, filter and powder-handling needs

Plant integration

Utilities, controls, cleaning and installation

Pilot work supports different materials and powder objectives



Pharmaceutical

Controlled moisture and repeatability



Food ingredients

Aroma, solubility and recovery



Botanical extracts

Stickiness and thermal sensitivity



Dairy and protein

Solubility and deposit behavior



Ceramics

Dispersion and particle control



Battery materials

Safety, moisture and particle targets

Start with the material information you already have



A useful first inquiry includes

- Material name and feed form
- Known solids, viscosity and solvent basis
- Target powder properties
- Available sample quantity
- Current project stage and expected capacity

Material Evaluation

<http://oltdryer/material-evaluation/>

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